MORPHO · RISK · INTERNAL

# Oct 10 Flash Crash Consequences - Incident Report

Date: Feb 23, 2026 · To: Renan & Luciano · From: Ivan Piton

P0 INCIDENT | Bad Debt Realized | Protocol Isolation: INTACT | MEV Capital - 2 Vaults Affected

Overview | A · Diagnosis | B · Actions | Curator Comms | Bad Debt Data | SQL Query

## 01 Overview

### INCIDENT SUMMARY

A macro-triggered flash crash on **Oct 10, 2025** (BTC -14%, Gas >500 gwei) caused cascading liquidations across DeFi. Morpho's market isolation architecture contained contagion at the protocol level, but **two MEV Capital curated USDC vaults** realized bad debt due to collateral depeg events in exotic stablecoin markets (xUSD and sdeUSD).

While there is not a direct causal effect between Oct 10 drop and bad debt that occured on these two MEV Capital vaults, it is clear that the general market uncertainty, and sudden price drops have broken various market structures including the strategies these vaults have previously relied on. This research assumes, with high confidence, that there are indirect links between the macro flash crash, leverage unwinds, and exotic stablecoin depegs.

Protocol functionality remained intact. Losses were socialized to vault depositors - no DAO or protocol-level funds were impacted.

**Arbitrum Vault Loss**

**~$700K**

xUSD (Stream Finance) · ~12% of vault exposure

**Ethereum Vault Loss**

**~3.6%**

sdeUSD (Elixir) · haircut to vault share price

**Est. Total Bad Debt**

**~$1.3M**

Combined across both vaults

**Protocol Isolation**

**HELD**

No contagion to other vaults or markets

### AFFECTED VAULTS

| CHAIN | VAULT ADDRESS | COLLATERAL | ROOT CAUSE | IMPACT |
|---|---|---|---|---|
| Arbitrum | 0xa606...019Cf | xUSD (Stream Finance) | Stream collapse → xUSD depeg to $0.30 → liquidation unprofitable | ~$700K absolute |
| Ethereum | (0xd630...3a3D) (0x0f95...b129) | sdeUSD (Elixir) | Elixir retired deUSD → 98% value loss → bad debt on-chain | ~3.6% share price haircut |

### KEY FINDING

Both failures share the same root cause: **exotic stablecoin collaterals that faced insufficient secondary market liquidity** were whitelisted in curator supply queues without adequate exit plans or liquidity. When the depegs occurred, liquidations were either unprofitable or impossible at any meaningful recovery rate.

Important, even though obvious, to note is that this is a curator risk parameter failure, not a protocol failure.

## A Diagnosis

### METRICS TO REVIEW DURING EVENT

| METRIC | SOURCE | WHAT TO LOOK FOR | CADENCE |
|---|---|---|---|
| **Oracle Heartbeats** | On-chain / Chainlink / Others | Stale price feed or reporting <$1.00 for collateral token | Real-time |

| METRIC | SOURCE | WHAT TO LOOK FOR | CADENCE |
|---|---|---|---|
| **badDebtAssets (Liquidate event)** | Dune / RPC | Non-zero value emitted - confirms bad debt realized | Real-time |
| **Vault Share Price** | convertToAssets(1e18) | Drop vs. previous block = realized loss socializing | Real-time |
| **Vault Utilization** | Morpho API / Dune | >95% utilization in healthy markets signals panic withdrawal crunch | Real-time |
| **Collateral Token Price** | DEX / Chainlink | Depeg below $0.98 for >5 blocks → P1 alert trigger | Real-time |
| **Liquidation Queue Depth** | On-chain | Positions eligible for liquidation but not yet taken = at-risk exposure | Near real-time |
| **Gas Price** | ETH Gas tracker | >300 gwei → liquidation bots may pause, increasing bad debt risk | Real-time |
| **Withdrawal Volume** | Morpho API | Spike in withdrawals → monitor for healthy market liquidity crunch | Near real-time |
| **BTC/ETH Price** | Tradingview feeds | Macro context: >10% drop → elevated DeFi stress event likely | Real-time |
| **L2 -> L1 Bridge Volume** | Dune | Inflows and outflows to/from L2s to Eth mainnet | Near real-time |

### REAL-TIME PRIORITY DASHBOARD (FIRST 30 MINUTES)

**Watch simultaneously:**

- **Oracle feed for affected collateral** - is xUSD / sdeUSD still reporting $1.00 or depegged? A stale oracle at $1.00 while spot is $0.30 means liquidators cannot profit → bad debt accumulates silently.
- **badDebtAssets in Liquidate events** - query Dune or query via RPC to MorphoBlue_evt_Liquidate filtered by vault's market IDs. Any non-zero value is a confirmed loss event.
- **convertToAssets(1e18) on vault contract** - this is the single clearest signal of realized bad debt in Vault V1. A drop vs. the prior day's value = loss has been socialized to depositors.
- **Withdrawal queue + utilization** - high gas + falling share price can trigger panic withdrawals. If utilization in healthy markets hits >95%,

new withdrawals will fail, compounding user panic.

### HOW THE BAD DEBT WAS LOCATED (DUNE)

The SQL query in the **SQL Query** section was used to scan `MorphoBlue_evt_Liquidate` events between Oct 10–Nov 15 filtering for `badDebtAssets > 0`. By joining against `MorphoBlue_evt_CreateMarket`, the collateral token address was extracted from `marketParams`, allowing identification of the specific xUSD and sdeUSD markets within the MEV Capital vault supply queues.

## B Actions

### RESPONSE TIMELINE

First 15 min

**Isolate · Quantify · Notify**

- **Isolate:** Confirm MEV Capital has removed xUSD and sdeUSD markets from vault supply queues. No new deposits should flow into bad debt markets. *Status on Oct 10: CONFIRMED.*
- **Quantify:** Run Dune query to confirm exact badDebtAssets values. Calculate % haircut per vault (ETH: 3.6%, ARB: ~$700K / vault TVL).
- **Notify:** Send P0 Telegram to MEV Capital ops. Alert internal Risk + Product channels. If share price hasn't yet updated on the front end, request curator to pause deposit UI to prevent new users entering at stale valuation.
- **Triage:** Confirm no contagion to other vaults - check that healthy markets (WBTC/USDC, wstETH/USDC) are unaffected.

**Coordinate · Decide · Communicate**

- **Internal brief:** Bring Product and Eng up to speed. Discuss whether "Asset Injection" (direct supply to market on vault's behalf to restore share price) is feasible and appropriate. Per Tutorials V1/Bad Debt, this is a valid recovery path.
- **Curator stance:** Confirm with MEV Capital: will they seek Asset Injection to restore share price, or socialize losses? Oct 10 outcome: socialized loss, coordinating off-chain recovery with Elixir/Stream Finance.
- **Draft depositor comms:** Prepare a clear, factual statement - what happened, why, what the loss is, and what is being done. Await MEV Capital approval before publishing.

First 60 min

- **Monitor withdrawal queue:** Do NOT pause withdrawals unless protocol safety is at risk. Exiting users should take their fair haircut based on current share price.

First 24 hrs

**Resolve · Publish · Stabilize**

- **Post-mortem published:** Acknowledge specific market failures (xUSD / sdeUSD) while clearly communicating that Market Isolation prevented broader contagion.
- **Withdrawals remain open:** Share price reflects bad debt. Users exiting now take their proportional haircut - this is correct Vault V1 behavior.
- **Off-chain recovery:** Track progress of MEV Capital's coordination with Elixir compensation program and Stream Finance. Report status to internal stakeholders.
- **Exec briefing sent:** Short summary email to leadership (see Curator Comms section) - TL;DR, impact, and recommendation.

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## FOLLOW-UP IMPROVEMENTS (1–2 WEEKS)

### Alerting (Internal)

Implement a new **P1 alert: Collateral Token < $0.98 for >5 consecutive blocks**. This gives a ~60 second warning before liquidations become unprofitable, enough time to notify curator and monitor escalation.

### Curator Guidance (External)

Mandate that all curators submit an **"Exit Plan"** for any localized or low-liquidity stablecoin collateral before it can be whitelisted. Exit plan must include: secondary market depth, oracle source, depeg contingency, and LTV rationale.

### Monitoring (Internal)

Add a **Dune dashboard tracking badDebtAssets > 0 events** across all Morpho Blue markets on Ethereum and Arbitrum, updated every 5-10 minutes, with webhook integration (e.g. Telegram).

### Onboarding (Process)

Introduce a **collateral due diligence checklist** for new market listings: liquidity depth on primary DEX, peg mechanism documentation, centralization risk score, and minimum 30-day track record requirement for novel stablecoins.

RECOMMENDATION: HANDLING THE BAD DEBT

- **Do not bail out with DAO funds** - this would create moral hazard for future curators. Losses must be socialized to vault depositors as designed.
- **Asset Injection path** - if MEV Capital chooses to restore share price (good faith recovery), they can supply assets directly to the market on the vault's behalf. This is supported by Morpho's V1 bad debt tutorials and is the cleanest UX outcome.
- **Withdrawals stay open** - preventing withdrawals would be more damaging than the loss itself. Trust in protocol mechanics must be preserved.
- **Off-chain coordination with Elixir/Stream** - pursue compensation programs aggressively but do not promise depositors recovery until terms are confirmed in writing.
- **Public comms framing** - lead with Market Isolation success, not the loss. The story is: "Morpho's design worked. Two exotic collaterals failed. The protocol and all other vaults are unaffected."

## 03 Curator Comms

TELEGRAM - MEV CAPITAL OPS (P0 ALERT)

TELEGRAM · SENT WITHIN 15 MIN OF DETECTION · MESSAGE 1

🚨 **URGENT / Bad Debt Realization - MEV Capital USDC Vault (Arbitrum)**

We've detected realized bad debt in your ARB USDC vault:
• **ARB Vault** (0xa606...019Cf) - ~$700K loss from xUSD (Stream Finance) depeg

**Action Required:**
1. Confirm removal of xUSD from supply queue immediately
2. Decide: Asset Injection to restore share price, or socialize losses?
3. Approve draft depositor comms - we need to publish before secondary social noise starts
4. Do you have a direct contact at Stream Finance for recovery coordination?

Protocol isolation held - no other vaults affected. Standing by for your confirmation.

TELEGRAM · SENT WITHIN 15 MIN OF DETECTION · MESSAGE 2

🚨 **URGENT / Bad Debt Realization - MEV Capital USDC Vault (Ethereum)**

We've detected a bad debt event in your ETH USDC vault:
• **ETH Vault** (0xd630...3a3D) - ~3.6% share price haircut from sdeUSD (Elixir) retirement. Elixir retired deUSD (declaring it worthless/no value) and sunset the token.

**Action Required:**
1. Confirm removal of sdeUSD from supply queue immediately
2. Decide: Asset Injection to restore share price, or socialize losses?
3. Approve draft depositor comms - we need to publish before secondary social noise starts
4. Do you have a direct contact at Elixir for recovery coordination?

Protocol isolation held - no other vaults affected. Standing by for your confirmation.

## INTERNAL EXECUTIVE BRIEFING - EMAIL

EMAIL · TO: EXEC / LEADERSHIP · SENT WITHIN 60 MIN · EMAIL 1

**Subject: Incident Summary - MEV Capital xUSD Bad Debt**

**TL;DR:** A flash crash triggered a depeg in xUSD (Stream Finance), used as collateral in the MEV Capital ARB USDC vault (0xa606...019Cf). Estimated loss ~$700K (~12% of exposure). Protocol functionality and market isolation are intact.

**Impact:**
• **Users:** ARB vault depositors face ~$700K absolute loss
• **Protocol:** No bad debt at protocol level - loss socialized to vault depositors as designed

• **Reputation:** MEV Capital marketed this vault as low-risk; xUSD collateral choice is now under scrutiny

**Recommendation:**
Support MEV Capital in off-chain recovery efforts (Stream Finance coordination) but do not commit DAO funds - this would set a moral hazard precedent. Curator risk decisions are the curator's responsibility. Focus external comms on Market Isolation success, not the loss quantum. Use this event to accelerate the collateral due diligence checklist rollout.

Full post-mortem to follow within 24 hours. Standing by for questions.

EMAIL · TO: EXEC / LEADERSHIP · SENT WITHIN 60 MIN · EMAIL 2

**Subject: Incident Summary - MEV Capital sdeUSD Bad Debt**

**TL;DR:** The retirement of sdeUSD (Elixir) caused a share price haircut in the MEV Capital ETH USDC vault (0xd630...3a3D). Estimated loss ~3.6% of share price. Protocol functionality and market isolation are intact.

**Impact:**
• **Users:** ETH vault depositors face a ~3.6% share price haircut
• **Protocol:** No bad debt at protocol level - loss socialized to vault depositors as designed
• **Reputation:** MEV Capital marketed this vault as low-risk; sdeUSD collateral choice is now under scrutiny

**Recommendation:**
Support MEV Capital in off-chain recovery efforts (Elixir compensation program) but do not commit DAO funds - this would set a moral hazard precedent. Curator risk decisions are the curator's responsibility. Focus external comms on Market Isolation success, not the loss quantum. Use this event to accelerate the collateral due diligence checklist rollout.

Full post-mortem to follow within 24 hours. Standing by for questions.

## INFORMATION FLOW BY STAKEHOLDER

| STAKEHOLDER | KEY INFORMATION TO SEND | TIMING |
|---|---|---|

| STAKEHOLDER | KEY INFORMATION TO SEND | TIMING |
|---|---|---|
| **MEV Capital (Curator)** | Exact vault addresses, bad debt amounts, queue removal confirmation, asset injection options, comms draft for approval | First 15 min |
| **Product / Eng** | Share price drop mechanism, potential need for asset injection tooling, front-end deposit pause feasibility | First 60 min |
| **User Support** | FAQ: what happened, are funds safe, will withdrawals work, what is the haircut, expected timeline for resolution | First 60 min |
| **Executives** | TL;DR loss size, protocol isolation status, curator accountability framing, comms strategy, no DAO bail-out stance | First 60 min |
| **Risk Committee** | Full technical breakdown: oracle failure, liquidation economics, market IDs, root cause per vault, improvement proposals | First 24 hrs |
| **Community / Twitter** | Post-mortem: incident summary, isolation success, curator accountability, no protocol-level impact, next steps | 24 hrs |

## 04 Bad Debt Data

### REALIZED BAD DEBT - AFFECTED MARKETS (OCT 10 – NOV 15 2025)

| MARKET | COLLATERAL TYPE | BAD DEBT (USD) | EVENT DATE | CHAIN |
|---|---|---|---|---|
| **sUSDe/USDC** | Ethena Synthetic Stablecoin | **$2.40M** | Nov 15 | Ethereum |
| **xUSD / sdeUSD** | Stream / Elixir Stablecoin | **~$700K–$1.0M** | Oct 10 | Arbitrum + ETH |

### BAD DEBT BY MARKET (NORMALIZED)

sUSDe / Ethena  Elixir

### RISK ASSESSMENT

🔴 **High** **sUSDe/USDC - Immediate Review Required**

$2.4M in a single Nov 15 cluster represents 55%+ of period losses. Ethena's sUSDe relies on delta-neutral funding; a funding rate reversal or redemption crunch causes rapid depeg. Recommend: review oracle freshness, LTV caps, consider supply cap reduction pending post-mortem.

🟠 **Medium-High** **xUSD / sdeUSD - Oct 10 Root Cause**

Direct cause of the Oct 10 incident. xUSD depegged to $0.30 (Stream Finance collapse); sdeUSD lost 98% of value (Elixir retirement). Both were low-liquidity, novel stablecoins without credible exit plans. These markets should be treated as a case study for mandatory collateral due diligence requirements going forward.

📋 **Systemic** **Cross-Market Pattern**

All losses trace to exotic or synthetic stablecoin collaterals - assets that appear stable but carry hidden liquidity and depeg risk. The protocol's liquidation infrastructure is tuned for volatile assets, not soft de-pegs where price moves are slow but deep.

## 05 SQL Query

### DUNE QUERY - REALIZED BAD DEBT EVENTS (OCT–NOV 2025)

**Purpose:** Identify bad debt events for MEV Capital USDC vaults during the crash window by scanning `MorphoBlue_evt_Liquidate` for non-zero `badDebtAssets`, joined with market creation events to extract collateral token addresses (used to filter for xUSD, sdeUSD, and related markets).

```sql
-- Realized Bad Debt on Morpho Blue (Oct–Nov 2025)
-- Filters for liquidations where badDebtAssets > 0
-- Joins CreateMarket to identify collateral token per market

WITH bad_debt_events AS (
  SELECT
    l.evt_block_time,
    l.evt_tx_hash,
    l.id                        AS market_id,
    l.badDebtAssets / 1e6       AS bad_debt_usdc_amount, -- USDC (6 decimals)
    l.badDebtShares,
    m.marketParams
  FROM morpho_blue_ethereum."MorphoBlue_evt_Liquidate" l
  LEFT JOIN morpho_blue_ethereum."MorphoBlue_evt_CreateMarket" m
    ON l.id = m.id
  WHERE l.evt_block_time >= TIMESTAMP '2025-10-10 00:00:00'
    AND l.evt_block_time <= TIMESTAMP '2025-11-15 23:59:59'
    AND l.badDebtAssets > 0
)

SELECT
  b.evt_block_time,
  b.market_id,
  json_extract_scalar(b.marketParams, '$.collateralToken') AS collateral_token,
  b.bad_debt_usdc_amount,
  b.evt_tx_hash
FROM bad_debt_events b
ORDER BY b.evt_block_time DESC;
```

### QUERY NOTES

| FIELD | NOTES |
|---|---|
| `badDebtAssets / 1e6` | Normalizes raw on-chain units to human-readable USDC amount (6 decimal token) |

| FIELD | NOTES |
| --- | --- |
| `badDebtAssets > 0` | Key filter - in a healthy liquidation this is zero. Non-zero = protocol absorbed a loss |
| `json_extract_scalar(...collateralToken)` | Pulls collateral address from marketParams JSON - use this to identify xUSD / sdeUSD markets |
| `LEFT JOIN CreateMarket` | Required to access marketParams - not available directly on the Liquidate event |
| `ORDER BY evt_block_time DESC` | Most recent events first - useful for triage during a live incident |